



3D culture of HepaRG cells in GelMa and its application to bioprinting of a multicellular hepatic model

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ARTICLE INFO

Keywords:

Bioprinting
Methacrylated gelatin
Hepatocyte
HepaRG
3D liver models

ABSTRACT

Bioprinting is an emergent technology that has already demonstrated the capacity to create complex and/or vascularized multicellular structures with defined and organized architectures, in a reproducible and high throughput way. Here, we present the implementation of a complex liver model by the development of a three-dimensional extrusion bioprinting process, including parameters for matrix polymerization of methacrylated gelatin, using two hepatic cell lines, Huh7 and HepaRG. The printed structures exhibited long-term viability (28 days), proliferative ability, a relevant hepatocyte phenotype and functions equivalent to or better than those of their 2D counterparts using standard DMSO treatment. This work served as a basis for the bioprinting of complex multicellular models associating the hepatic parenchymal cells, HepaRG, with stellate cells (LX-2) and endothelial cells (HUVECs), able of colonizing the surface of the structure and thus recreating a pseudo endothelial barrier. When bioprinted in 3D monocultures, LX-2 expression was modulated by TGFβ-1 toward the induction of myofibroblastic genes such as ACTA2 and COL1A1. In 3D multicellular bioprinted structures comprising HepaRG, LX-2 and endothelial cells, we evidenced fibrillar collagen deposition, which is never observed in monocultures of either HepaRG or LX-2 alone. These observations indicate that a precise control of cellular communication is required to recapitulate key steps of fibrogenesis. Bioprinted 3D co-cultures therefore open up new perspectives in studying the molecular and cellular basis of fibrosis development and provide better access to potential inducers and inhibitors of collagen expression and deposition.

1. Introduction

The liver is the organ responsible for the biotransformation and elimination of drugs and xenobiotics. Therefore, developing *in vitro* assays based on human hepatic cells are one of the most critical steps for assessing the metabolism and/or toxicity of drugs or exogenous compounds [1].

These last decades, conventional two-dimensional (2D) cell cultures of hepatic model systems have been extensively developed by many laboratories. However, murine and human normal hepatocytes do not

survive more than one week in 2D culture and rapidly lose their differentiated functions [2,3] whereas transformed human hepatocytes derived from primary human hepatocytes do not, most of the time, recapitulate the differentiation status of native parenchymal cells [4,5]. Moreover, 2D cell culture of hepatic model systems does not reliably recapitulate the liver structure, phenotype and multicellular architecture, leading to a gap between observed *in vitro* toxicity and *in vivo* human hepatotoxicity [6]. Animal models have been the tool of choice for assessing the preclinical safety of new drug candidates and are still required by legislation. However, it is now recognized that the results

Abbreviations: 2D, two-dimensions; 3D, three-dimensions; 3 MC, 3-methylcholanthrene; CYP, Cytochromes P450; CDDP, cisplatin; DMSO, dimethyl sulfoxide; ECM, extracellular matrix; EdU, 5-ethynyl-2'-deoxyuridine; EROD, Ethoxyresorufin-O-deethylation GelMa; methacrylated gelatin, HSC; hepatic stellate cells, LAP; lithium phenyl-2,4,6 trimethylbenzoylphosphinate, MROD; Methyresorufin-O-deethylation, TGFβ-1; Transforming growth factor beta 1, TPEF; two-photon excitation microscopy, SHG second harmonic generation.

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<https://doi.org/10.1016/j.biomaterials.2020.120611>

Received 9 March 2020; Received in revised form 24 November 2020; Accepted 14 December 2020

Available online 16 December 2020

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obtained on animal models only partially recapitulate toxicity observed in humans and the specificities of the biotransformation of drugs have been largely described [7]. Besides, ethical considerations related to the utilization of these models tend to limit their use.

To develop a physiological human liver model, the pharmaceutical industry is interested in developing an ideal *in vitro* model combining both the benefits of 2D and 3D cultures. Conventional 2D cultures, whether of primary, immortalized or stem cells derived hepatocytes, is still mainly used today during the first stage of biotransformation and hepatic toxicity studies. Easily useable, these controlled and standardized culture conditions make them good models for drug screening and to high-throughput testing. In recent decades, the 3D culture of hepatic cells has been able to overcome the weaknesses of 2D cultures providing, for example, a matrix environment that mimics the mechanical cues from hepatocyte microenvironment and promotes cell-cell interactions [8–10]. 3D monoculture and co-culture of hepatic cells improve the viability and the functional capability of hepatocytes compared to 2D monolayer cultures.

The main prerequisite for developing an ideal 3D liver model is the ability to co-cultivate parenchymal and non-parenchymal cells as to mimic intercellular events and cross-talk [11–13], and to mimic the complexity and diversity of metabolism and drug toxicity pathways [14]. The cells must remain viable and functional for several weeks in culture, which will allow studies long enough to reflect repeated low doses exposure as *in vivo*. The phenotype of the liver cellular model should be as close as possible to the *in vivo* phenotype, in particular the activity of phase I and II biotransformation enzymes and transporters, major actors in the metabolism of drugs. The model must be reproducible, robust and allow high throughput screening of the drug or compounds.

3D bioprinting offers an attractive alternative to classic 3D models [15,16]. The controlled and accurate cell deposition by the bioprinter provides 3D models with defined microarchitecture and the possibility to recreate vascularized structures [17,18]. So far, very few efficient bioprinted hepatic models have been developed. In most of these models, long-term viability is not assured and few studies have evaluated the functionality of the printed cells or the toxicity and biotransformation of exogenous compounds [18–21]. Bioprinting of murine hepatocytes has led to promising models in terms of long-term viability [22,23] but in addition to the ethical and commercial aspect, studies carried out on primary murine hepatocytes are limited by significant differences in metabolism and pharmacokinetics compared to human hepatocytes. One of the most successful and described bioprinted models was developed by the Organovo team. A primary human hepatocyte paste was co-printed with human stellate and HUVECs cells, fabricating a functional model of human liver tissue [24,25]. Unfortunately, the cost and difficulty of obtaining freshly isolated primary cells and the use of homemade extrusion printers do not allow this model to be routinely replicated and to be applied to high throughput studies. We have developed here a cheap and easily useable 3D model using an Allevi commercial extruder. The selected ink, gelatin methacrylate (GelMa), is easily accessible or synthesizable and is widely used in extrusion-based systems [26,27]. We have optimized the printing conditions to obtain a good structural integrity as well as an optimal viability and cell differentiation over the long term.

The HepaRG cell line, isolated from hepatocarcinoma cells of a hepatitis B virus infected patient [28], is currently the gold standard for transformed human hepatocytes. HepaRG cells retains bipotent hepatic progenitor-like characteristics and can partially recapitulate the liver phenotype when they are cultivated in the presence of DMSO at 1.7% to 2% (v/v) in 2D [28–30]. 3D models of the HepaRG cell line have been showed to be an attractive tool for toxicological studies showing higher cell differentiation than in 2D culture [31,32]. The HepaRG cell line, the “gold standard” for liver tests, and the LX-2 cell line, derived from activated human stellate cells [33], were successfully printed in a GelMa matrix to obtain a complex and functional 3D model. Not only the

structures were shown to be viable for at least one month, but we have, for the first time, demonstrated the ability of the bioprinted HepaRG cells to differentiate and proliferate in a 3D environment in absence of DMSO.

The HepaRG cells were then used to mimic physiopathological phenomenon, hepatic fibrosis. *In vivo*, following chronic attacks, the secretion by hepatocytes and Kupffer cells of pro-fibrosing signals such as TGF β -1 causes activation of quiescent hepatic stellate cells (HSC) which acquire a myofibroblastic phenotype. They become proliferating, contractile and activate tissue healing by synthesizing components of the extracellular matrix (ECM) such as type I collagen. The *in vitro* study of mechanisms of development of hepatic fibrosis such as activation and matrix remodeling by stellate cells requires the establishment of co-cultures between, inter alia, hepatic cells and hepatocytes [12,13,34,35]. However, the culture of stellate cells remains today a technological obstacle in obtaining relevant hepatic fibrosis models. Primary HSCs, human or murine, are widely used but, when cultivated in a monolayer on rigid supports (such as plastic culture dishes), self-activate in 7–10 days from a quiescent to a myofibroblastic phenotype [36]. We have demonstrated that, in our 3D system, LX-2 cells exhibit features of quiescent stellate cells and initiate an activation process in response to TGF β -1 treatment. Moreover, we evidenced a deposition of fibrillar collagen when LX-2 cells were co-cultured with HepaRG and endothelial cells. These observations make our 3D system a promising model for liver fibrosis and open up new perspectives in studying the mechanisms of fibrogenesis, and the effects of exogenous factors on this process.

2. Material and methods

2.1. Reagents

DMEM (41966-029) and William's E (22551-022) mediums were from Gibco (ThermoFisher, Waltham, MA USA). Lipofectamine RNAi max transfection reagent and TRIzol were from Invitrogen (Invitrogen, Carlsbad, CA, USA). Fetal bovine serum (FBS) and sucrose were purchased from Eurobio (Evry, France). Penicillin/Streptomycin and FBS HyClone III were from Thermo Fisher (Waltham, MA USA), WST1 reagent, geneticin, Human insulin, hydrocortisone hemisuccinate, DMSO (D4540), formaldehyde, gelatin, 5-ethynyl-2'-deoxyuridine, salicylamide, CY5-azide, ethoxyresorufin and methoxyresorufin, 3- methylcholanthrene (3 MC) and rifampicine were from Sigma-Aldrich (St. Louis, MO, USA). Methacrylated gelatin (GelMa) was synthesized by the ART BioPrint (Bordeaux, FR). Lithium phenyl 2,4,6 trimethylbenzoylphosphinate (LAP) was from TCI (Japan) and U0126 was purchased from Promega (Mannheim, GE). Phenobarbital was from the Coopération pharmaceutique française. The Dapi Fluoromount-G was purchased from Southern Biotech (Birmingham, AL, USA). The EBMTM-2 Endothelial Cell Growth basal medium and EGMTM-2 BulletKitTM were from LONZA (Aubergenville, France).

2.2. Cell cultures

Huh7 cells (ECACC 01042712) were obtained from Health Protection Agency Culture Collections, Salisbury, UK. DNA transfections were carried out using Lipofectamine RNAi max transfection reagent according to the manufacturer's instructions. Briefly, cells were plated in a 35 mm-petri dish. 3 μ L EmGFP-geneticin vector and 5 μ L of Lipofectamine RNAi max were mixed in 200 μ L Opti-MEM[®] and then added to the cells. After overnight incubation, cells were incubated 24h in DMEM medium for 24h, then selected by geneticin treatment at 10 mg.mL⁻¹ during 48h. After selection, Huh7 GFP + cells were cultured in DMEM containing 4.5 g/L glucose and pyruvate supplemented with 10% (v/v) FBS (EuroBio), 100 units.mL⁻¹ penicillin and 100 μ g mL⁻¹ streptomycin, 2 mM l-glutamine and 500 μ g mL⁻¹ geneticin.

LX-2 were provided by Dr N. Theret (Irset, France) and were cultured in DMEM containing 4.5 g/L glucose and pyruvate supplemented with

10% (v/v) fetal bovine serum (EuroBio), 100 units.mL⁻¹ penicillin, 100 µg.mL⁻¹ streptomycin and 2 mM L-glutamine.

GFP-HUVECs (ZHC-2402, CellWorks, Buckingham, UK) were cultured in a mix of EBM™-2 Endothelial Cell Growth basal medium and William's E medium (1:1) supplemented with 10% (v/v) FBS Hyclone III, 100 units.mL⁻¹ penicillin, 100 µg.mL⁻¹ streptomycin, 5 µg mL⁻¹ human insulin, 2 mM L-glutamine, and 5×10^{-5} M hydrocortisone hemisuccinate. hFGF-B, VEGF, Ascorbic Acid, hEGF, GA-1000 and Heparin issued from the EGM™-2 BulletKit™ were added to the medium at concentrations recommended by the supplier.

Undifferentiated HepaRG cells (Biopredic, Saint Grégoire, France) were cultured in William's E medium without L-Glutamine supplemented with 10% (v/v) fetal bovine serum Hyclone III, 100 units.mL⁻¹ penicillin, 100 µg mL⁻¹ streptomycin, 5 µg mL⁻¹ human insulin, 2 mM L-glutamine, and 5×10^{-5} M hydrocortisone hemisuccinate for 14 days. Hepatic differentiation in 2D culture was induced by treatment with 2% (v/v) DMSO for an additional 14 days. Undifferentiated HepaRG cells were cultivated on low attachment plate for 15h and the cluster of cells were harvested for mixing with GelMa before bio-printing (See patents n° EP2018030560320180516/WO2019219828). Primary human hepatocytes (PHH) were obtained from the processing of biological samples through the Centre de Ressources Biologiques (CRB) Santé of Rennes BB-0033-00056. The research protocol was conducted under French legal guidelines and fulfilled the requirements of the local institutional ethics committee. PHH were isolated by a two-step collagenase perfusion procedure [37] and immediately proceeded.

2.3. Bio-ink preparation and bioprinting

Methacrylated gelatin was synthesized as previously described [26]. Briefly, a 10% (w/v) gelatin (Type A, 300 bloom from porcine skin, Sigma) solution, in PBS, was warmed at 60 °C for 2 h under stirring. Then, the temperature was set to 50 °C and 0.14 mL of methacrylic anhydride (Sigma) per gram of gelatin was added drop-wise. The solution was left to react for 4 h, under stirring, then diluted to 5% (w/v) in PBS. The reacted gelatin was precipitated using a 4-fold excess volume of cold acetone. The precipitated gelatin was recovered, vacuum dried for 30 min, and then redissolved at 10% (w/v) in PBS at 40 °C. It was then placed inside a 12–14 kDa molecular weight cutoff (MWCO) dialysis tubing (Sigma), and dialyzed against deionized water for 3 days, with two daily water changes. The purified GelMa was frozen at –80 °C, freeze dried, and stored at –20 °C. Methacrylation degree was $60.4 \pm 1.4\%$ (n = 3), as determined by TNBS assay. GelMa was then dissolved in Williams' E medium at 37 °C overnight. The bio-ink was then mixed to Lithium phenyl-2,4,6 trimethylbenzoylphosphinate dissolved in PBS (LAP, 10 mg.mL⁻¹) and to the cells. The final cell-laden bio ink was composed of 5% w/v GelMa, 0.3 or 0.1% w/v LAP as indicated, 0.5×10^6 /mL Huh7 or LX-2 cells or 2×10^6 /mL undifferentiated HepaRG cells. The bioprinted structures for the study of HepaRG alone have an estimated volume of 100 µL, encapsulating approximately 0.2×10^6 cells each. For the co-cultures, the bioprinted structure has a volume for HepaRG cells of approximately 150 µL (0.3×10^6 cells each) and LX-2 of 65 µL (0.0325×10^6 cells each); thus, giving a 10:1 cell ratio.

GelMa gels were characterized in terms of rheology. Briefly, photopolymerized gels, at 2.5, 5, 7.5 and 10% (w/v) were soaked for 24 h in PBS and then loaded in a Kinexus pro⁺ rheometer (Malvern Instruments, UK). Frequency sweeps were performed at a constant strain (0.1%) in an angular frequency range of 0.1–100 rad/s, at 37 °C.

For the bioprinting process, 100 µL of the bio-ink was cooled at 15 °C and extruded through a 23G needle at 1.4–2.4 bar, at 480 mm/min, using the extrusion printer Allevi 2 (Allevi, Philadelphia, PA, USA) in 48 wells plate. The 3D construct was designed using the software OnShape (Cambridge, MA, USA). The printed constructs were exposed to violet light (405 nm, 7 mW/cm²) during specified times to induce the radical polymerization of GelMa. They were then submerged in Williams' E medium with supplements and incubated at 37 °C and 5% CO₂. For

HUVECs seeding of constructs, the bioprinted HepaRG or HepaRG/LX-2 structures were cultured during a week, before being put in a low attachment plate containing a 1×10^6 /mL HUVECs suspension.

2.4. Cell viability

Cell viability assays were performed by using WST1 proliferation assay on cell-laden structures at the different days or times of culture. Briefly, structures were treated by 100 µL of WST1 reagent and put 30 min at 37 °C. Absorbance was read at 440 nm in a microplate reader (SpectrostarNano, BMG labtech, Champigny s/Marne, FR) and the values were converted into percentage of absorbance by comparison with controls at time 0. Measurement of LDH was performed on supernatants with the LDH-Glo™ Cytotoxicity Assay (Promega) by following manufacturer's instructions.

2.5. Proliferation quantification

For EdU incorporation, cell-laden constructs were treated 48h with EdU at 1 µM. For negative controls, cells were treated 24h prior EdU treatment by U126 at 20 µM until the fixation of the structures. Constructs were fixed and cryosectioned as described in the *Immunostaining procedure* section. EdU revelation was carried by performing permeabilization using 0.1% (v/v) Triton X-100 for 10 min. Then, samples were treated with a mix of CuSO₄ 1(µM), Tris pH 8.5 (0.1 M), ascorbic acid (0.1 M) and CY5-azide (1.5 µM, Sigma) for 1h. Cells were mounted with Dapi Fluoromount-G and the nucleus staining was monitored using a fluorescence microscope (Eclipse Ni-E, Nikon, Amsterdam, NL). Image processing was carried out using ImageJ software (<http://imagej.nih.gov/ij/>).

2.6. Immunostaining

Cell-laden structures were fixed in 4% formaldehyde, washed 3x in PBS and stored at 4 °C until use or immediately suspended in a 0.12 M phosphate buffer containing 10% w/v sucrose and incubated overnight. They were then embedded in a protective matrix composed of 7.5% w/v gelatin and 10% w/v sucrose, then frozen in isopentane and cryosectioned at a thickness of 7 µm. Slides were stained on the Discovery Automated IHC stainer using the discovery Rhodamin kit (Ventana Medical Systems, Tucson, AZ, USA). The concentrations and references of used primary antibodies can be found in [Supplemental Table 1](#). After rinsing, signal enhancement was performed using the Ventana Rhodamin kit and secondary antibody anti-rabbit HRP (760–4311, Roche) or secondary antibody anti-mouse HRP (760–4310, Roche) for 16 min of incubation. After removal of the instrument, slides were manually rinsed, stained when required with albumin antibody then a secondary antibody for albumin detection (Donkey anti goat 488), and mounted with Dapi Fluoromount-G.

2.7. RNA extraction and quantitative real time PCR

After 1, 2 or 4 weeks of culture, cell-laden structures were harvested, washed twice in PBS and total RNAs were extracted using TRIzol. The concentration of total RNAs were measured with a NanoDrop ND-1000 (NanoDrop Technologies). cDNA were synthesized using the High Capacity cDNA reverse transcriptase kit (Applied Biosystems Foster City, CA, USA), and Real-time PCR was performed using the Power SYBR Green PCR master mix (Applied Biosystems Foster City, CA, USA). Primer-BLAST (NCBI USA), was used to design primer sequences which were purchased from Eurogentec (Searing, BE). All used primers are listed in [Supplemental Table 2](#). The relative amount of measured mRNA in samples was determined using the $2^{-\Delta\Delta CT}$ method where $\Delta\Delta CT = (CT_{\text{target}} - CT_{\text{GAPDH}})_{\text{sample}} - (CT_{\text{target}} - CT_{\text{GAPDH}})_{\text{calibrator}}$. Results were expressed as the n-fold difference of target gene expression in samples as compared with the mean expression values of the target gene.

2.8. Transcriptomic analysis

Total RNAs were purified from freshly isolated human hepatocytes (PHH T0) (n = 5), from differentiated HepaRG after 14 days of culture on 2D (HRG 2D DMSO) (n = 5) and at day 14 of culture on GelMa (HepaRG 3D GelMa) (n = 4). The samples were concentrated using the RNA Clean & Concentrator-5 (Zymo, Irvine, CA, USA) and checked for RNA degradation based on the RNA Integrity Number (RIN > 6). 3' sequencing RNA Profiling (3' SRP-seq) libraries were made at the GenoBIRD facility of Nantes, France, and sequenced using the HiSeq 2500 (Illumina, San Diego, CA, USA) following the described protocol [38]. The analysis of the generated data was performed using R packages. The differentially expressed genes (FC > 2, p < 0.05) between HRG 2D DMSO and HRG GelMa D14 were functionally analyzed by computing enrichments for gene ontology (GO) terms, using the WEB-based database GENE SeT Analysis Toolkit (WebGestalt) restricted to protein coding data set. The presented data are selected as the top ten of the enriched categories, sorted by increasing enrichment ratios (FDR < 0.05).

2.9. Metabolic analysis

Cell culture medium samples were taken for the determination of albumin and urea content every 48h. Secreted albumin and pro-Coll1a1 were quantified using the human serum albumin Duoset Enzyme Linked Immunosorbent Assay (ELISA) and the human pro-Coll1a1 Duoset Enzyme Linked Immunosorbent Assay kits (R&D systems, Minneapolis, MN, USA), according to the manufacturer's instructions. Quantification was performed by measuring absorbance at 450 nm using a microplate reader (SpectrostarNano, BMG labtech, Champigny s/Marne, FR). Urea secretion in the medium was measured using a ChromaDazzle Urea Assay kit (AssayGenie, Dublin, IR) according to manufacturer's protocol, using medium as standard and reading the absorbance at 420 nm (SpectrostarNano, BMG labtech, Champigny s/Marne, FR).

2.10. CYP activities measurement by fluorescence

After a 24H induction with DMSO 0.1% v/v or 5 μ M 3 MC, Ethoxyresorufin-O-deethylation (EROD) was used as measurement of CYP1A activity and Methoxyresorufin-O-deethylation (MROD) for the CYP1A2 activity. EROD and MROD assays were performed as previously described [39]. The cells were washed with PBS and salicylamide (1.5 mM) was added to block phase II conjugation enzymes. 7-ethoxyresorufine or 7-methoxyresorufine was then added and the oxidation of the substrate was measured by fluorescent detection during 20 min, at 37 °C (SpectraMax GeminiXS, Molecular devices, San Jose, CA, USA). After the lecture, a WST1 test was performed to normalize activity with the quantity of viable cells.

2.11. CYP activities measurement by luminescence

Cytochrome P450 oxidase 2B6 and 3A4 activities were assessed by treating cells with DMSO 0.1% (v/v), or activating them by phenobarbital (0.2 mM, 72h) (CYP2B6) or rifampicine (5 μ M, 72h) (CYP3A4). Using the P450 GloAssay (Promega) according to the manufacturer's instructions, a CYP specific substrate Luciferin was added on cell-laden 3D constructs and incubated at 37 °C, 5% CO₂, for 1 h. Then, CYP-mediated conversion of Luciferin substrate to luciferin was determined. The supernatant was incubated with Luciferin detection reagent during 20 min at RT and the luminescence was measured. After the lecture, a WST1 test was performed to normalize activity with the quantity of viable cells.

2.12. SHG/TPEF microscopy

TPEF/SHG microscopy imaging was performed on MRic facility of

Biosit, University of Rennes1 (France). The SHG imaging system is composed of a confocal TCS SP5 scanning head (Leica Microsystems, Mannheim, Germany) mounted on a DMIRE2 inverted microscope (Leica Microsystems) and equipped with a MAITAI femtosecond laser (Spectra Physics, Santa Clara, CA). 60 $\times$ water immersion objective (Olympus LUMFL 60W x 1.1NA) was used for applying an 820 nm excitation to the sample. The SHG signal was collected in the forward direction using the condenser (S1, NA ¼ 0.9–1.4; Leica Microsystems), and the TPEF was epi-collected in the backward direction. IRSP 715 bandpass and 410 nm infrared (IR) filters (10 nm full width at half-maximum, FWHM) were placed before the photomultiplier tube. The image processing was performed with ImageJ software (National Institutes of Health).

2.13. Statistical analysis

Unless clearly specified in caption, all variables are expressed as means $\pm$ standard deviations (SDs) of at least 3 experimentations with 3 technical replicates. Evaluation of the difference between the mean values in each group was performed using *t*-test or one-way analysis of variance (ANOVA) (GraphPad Prism 6, GraphPad Software, Inc.; La Jolla, CA, USA). Significant differences are represented for the following p-value thresholds: p $\leq$ 0.05 (*/#); p $\leq$ 0.01 (**/##); p $\leq$ 0.001 (***), p $\leq$ 0.0001 (****).

3. Results

3.1. Physical properties of GelMa and Huh7 viability after bioprinting

We first examined the polymerization and printability of various matrices that can be used for printing by an extrusion process (Supplemental Table 3). Three criteria were assessed for the selection of the matrix: 1/its ability to polymerize and give a gel with sufficient rigidity to potentially maintain the structural and geometric integrity of the post-printing structures (polymerization) 2/the polymerized gel had to be viscous enough to complete the extrusion process satisfactorily, with a regular filament shape during printing and maintaining the structure's architecture during printing (extrudability) 3/the matrix should promote long-term culture of incorporated cells (cell culture). In view of our previous work on the incorporation and 3D culture of hepatic cells into collagen I matrix [8], we tried to bioprint it, as well as methacrylated collagen (ColMa), by adding, or not, polymerizing adjuvants (Phytigel or carrageenan, two plant based gelling agents; methacrylated hyaluronic acid; collagen I) to compensate for their low viscosity. Unfortunately, none of these mixes made it possible to obtain sufficient viscosity to pass the 2nd criteria, i.e. the ability to form gels viscous enough to withstand the extrusion process.

We then turned to methacrylated gelatin, a matrix widely used in extrusion bioprinting due to its suitable mechanical properties and biocompatibility. Due to the thermo-induced gelling capacity of gelatin, GelMa (5% w/v) is able to polymerize at low temperature (15 °C) to form a gel whose viscosity is sufficient to withstand the extrusion process (Criteria 1 and 3). To ensure optimal mechanical conditions of the GelMa promoting 3D hepatic cell culture, the physical properties of different GelMa compositions were analyzed. First, we examined the polymerization and printing properties of the gels as a function of the GelMa and LAP concentrations and different photopolymerization exposure times at 405 nm. At 20 °C, a 2.5% concentration of GelMa was not sufficient to ensure polymerization and consequently an optimal viscosity for the printing process (Fig. 1A). A minimum concentration of LAP and an optimized time of light exposure have been defined to reduce the phototoxicity and the production of excess free radicals [40] (Fig. 1B) while maintaining sufficient structural integrity of the printed GelMa for 28 days at 37 °C. Decreasing LAP concentration led to a diminished rate of cross-linking, and therefore to an altered structural integrity of the 3D bioprinted structures over time. Comparison of the

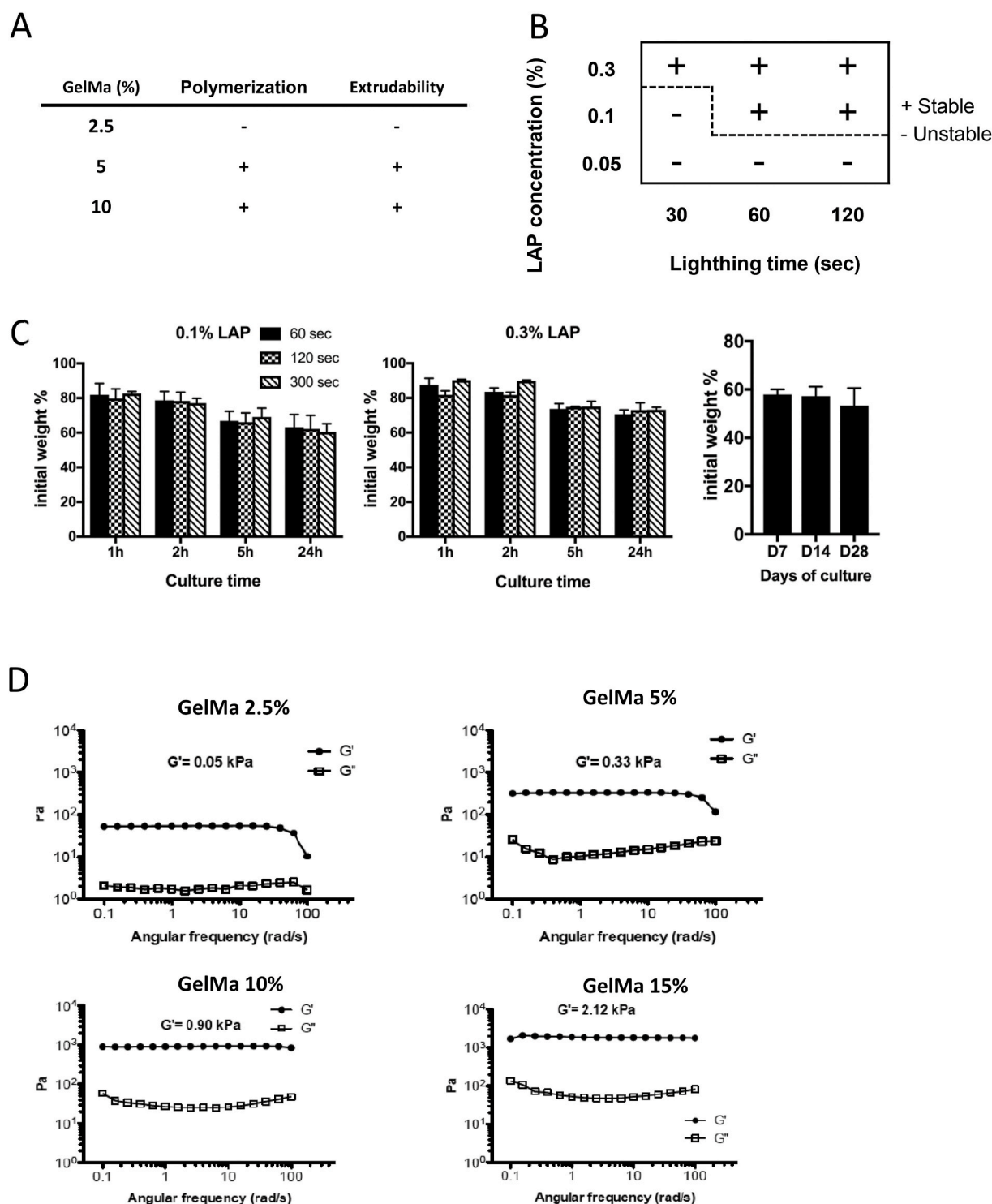


Fig. 1. Polymerization, printability and physical properties of GelMa. (A) Polymerization and extrudability at different concentrations: 2.5%, 5% and 10% of GelMa. (B) Structural integrity at 37 °C of biprinted models with 5% GelMa, 1 h after cross-linking at different LAP concentrations (%) and lighting times (sec). (C) Degradation kinetic of a 5% GelMa structure for 24 h (left) or 28 days (right) of culture at 37 °C after biprinting and cross-linking, depending on the LAP concentration and lighting duration. Results are presented as mean $\pm$ SD of $n = 3$ experiments. (D) Gel stiffness measured by rheology: elastic (G') and viscous (G'') modulus of GelMa at different concentrations: 2.5%, 5%, 10% and 15% (w/v).

weight of the gels obtained by polymerization in the presence of LAP concentrations at 0.1% or 0.3% showed a rapid decrease up to 40%, 5 h after the polymerization process. Then, a constant weight for at least 28 days indicated good stability of the gel in the culture medium at 37 °C (Fig. 1C). As expected, rigidity, analyzed by rheology (Fig. 1D), highly increased with the gelatin concentrations, consistent with other reports

showing that the mechanical properties of the gelatin scaffolds were highly dependent on gel concentrations and cross-linking processes [41]. A direct proportional relation was found between gelatin concentrations and the elastic (G') and viscous modulus (G'') with an exponential increased of G' from 0.05 to 2.12 kPa when the percentage of gelatin increased from 2.5% to 15%, respectively.

Next, we examined the short- and long-term viability of a hepatic transformed cell line, the Huh7 cells in GelMa before and after bioprinting and cross-linking (GelMa 5%, LAP 0.1%, 1 min of light exposure). No decrease in Huh7 cell metabolism (WST1 activity) could be observed after 4 h of bioprinting indicating that neither the GelMa, the cross-linking nor bioprinting processes were toxic at short times after

seeding (Fig. 2A). Interestingly, for minimal LAP concentration (0.1%) and lighting times (1 min), the WST1 activity increased until day 14 and then remained constant thereafter at least until 28 days (Fig. 2B and C) showing that cells could survive and/or proliferate in the matrix after bioprinting. Furthermore, we showed that the cells were able to divide in GelMa since Ki67 positive cells could be detected at days 5 (12%), 10

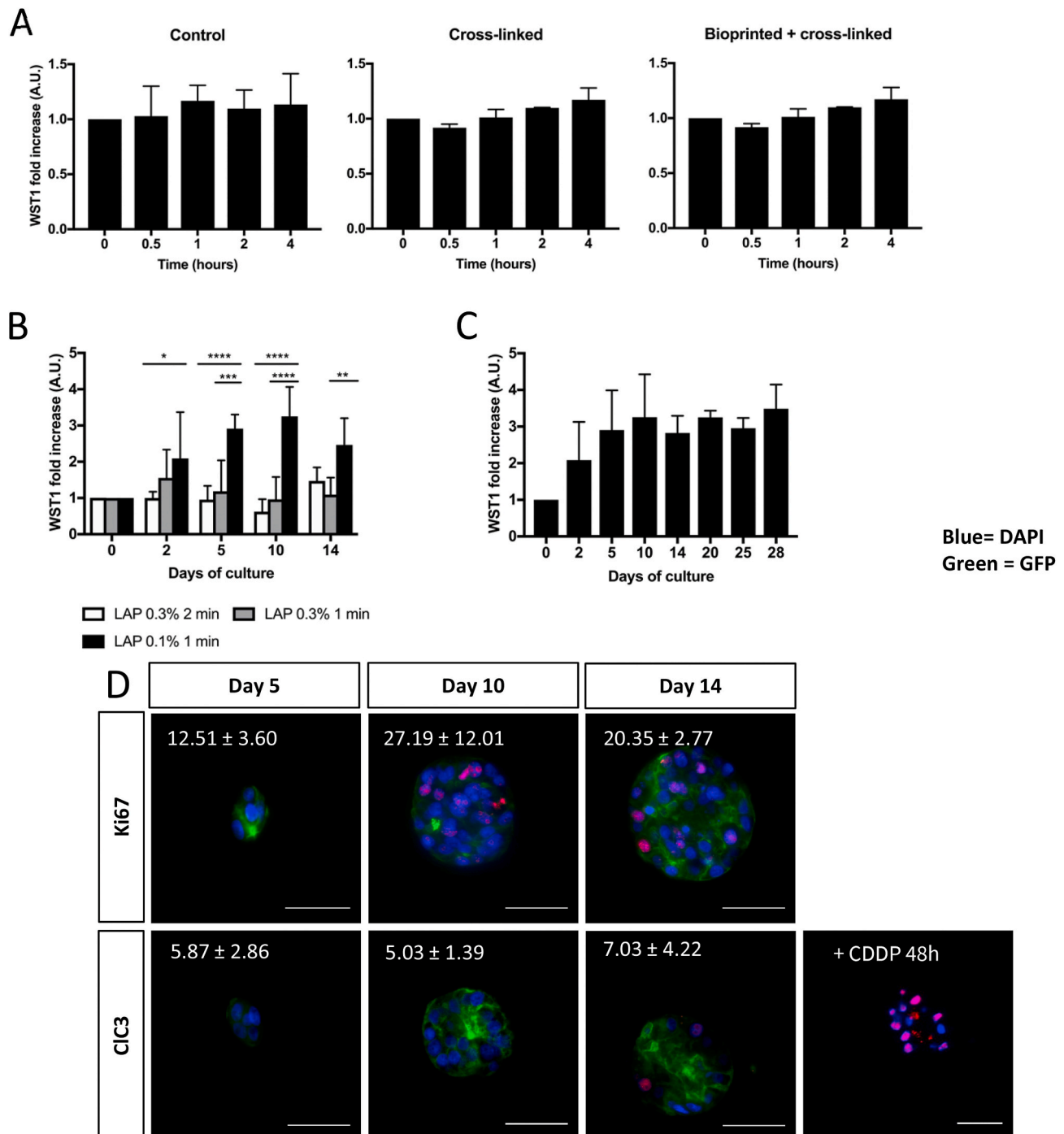


Fig. 2. Viability and cluster organization of the Huh7 cells in 5% GelMa. (A) Short-term viability of Huh7 cells in 5% GelMa, LAP 0.1%, before (control), after 1 min cross-linking, and after bioprinting and 1 min cross-linking at the indicated times (hours) after seeding. (B) Viability of Huh7 cells for 15 days after cross-linking with LAP at 0.1% or 0.3% and after 1 or 2 min cross-linking. (C) Long-term viability of Huh7 cells in 5% GelMa, LAP 0.1%, 1 min cross-linking, for 28 days of culture. (A–C) Viability data are represented as fold increase of WST1 activities compared to that of control (Day 0). Significance levels: * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$; **** $p \leq 0.0001$. (D) Cluster organization, proliferation and apoptosis of Huh7-GFP in 5% GelMa at 5, 10 and 14 days after seeding: Blue = DAPI, Green = GFP, Red = Ki67 or cleaved caspase 3 (CIC3). Scale bar = 50 μ m. Caspase positive control at day 15 after 20 μ M cisplatin treatment for 48 h (right image); quantifications of the % of Ki67 and cleaved caspase 3 = positive cells/total nucleus. All results are presented as mean $\pm$ SD of $n = 3$ experiments. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

(27%) and 14 (20%) (Fig. 2D), in accordance with the viability results described above (Fig. 2B and C). None or very few cleaved caspase-3 positive cells could be observed in these cultures 5, 10 and 14 days after seeding, whereas the cells responded well to an apoptotic inducer (CDDP) treatment at day 14 (Fig. 2D), indicating that cell death was very limited in these bioprinted structures. Based on these observations, we chose to work with GelMa and LAP concentrations of 5% (w/v) and 0.1% (w/v), respectively, and after 60 s of light exposure (at 405 nm).

3.2. Cluster organization, polarization and proliferation of HepaRG in GelMa

We applied this optimized bioprinting protocol to the gold standard human hepatic transformed cells, the HepaRG cell line. Undifferentiated HepaRG cells were either briefly (15h) cultivated in low-attachment plate in order to recreate partial cells-cells contact and bioprinted in a 5% GelMa, LAP 0.1% matrix, or left in 2D and fully differentiated by 2% DMSO (Fig. 3A). In absence of these cells interactions enhancement, the HepaRG cells cannot fully differentiate in GelMa (data not showed). The 3D cultures were carried out in the absence of DMSO in order to avoid the non-specific induction of some CYP P450, i.e. the CYP3A4 and CYP2B6, which has been previously reported in DMSO-induced cultures of HepaRG [29]. HepaRG developed in small clusters of cells 7 days after seeding in the 3D GelMa hydrogel (Fig. 3B). The cells appeared polarized as evidenced by the localization of E- and N-cadherins at the apico and lateral membranes. The hepatobiliary excretion transporter MRP2 located exclusively at apico/canalicular areas, confirming the expected polarization of the cells in the clusters all over the culture time (Day 7, 14 and 28). The epithelial phenotype of these cells was assessed by measuring the expression of mesenchymal and epithelial markers. We found that *CDH2*/N-Cadherin (a mesenchymal marker) was less expressed in 3D hydrogels compared to 2D DMSO cultures while *CDH1*/E-Cadherin (an epithelial marker) and *ABCC2*/MRP2 (an hepatocyte marker), poorly expresses in undifferentiated 2D HepaRG (2D) were expressed at the same level in the two cultures.

Then, we evaluated the proliferation potential of the HepaRG cells in 3D by studying the expressions of cell division markers (Cyclin D1 and Ki67), as well as the incorporation of EdU, a readout of DNA replication. Only small differences in the kinetics of expressions of these proliferation markers could be observed (Fig. 3C). The late G1 and S phase markers, cyclin D1 and Ki67 respectively, were expressed in 10–20% of the cells, depending on the culture times. The same levels of EdU positive cells were detected in parallel to the Ki67 expression kinetic. The nuclear incorporation of EdU was largely abrogated by the MAP kinase inhibitor U0126, indicating that HepaRG proliferation in 3D was MEK1/2-ERK1/2 dependent (Fig. 3D).

3.3. Expression of differentiation and biotransformation markers

To assess hepatic functions, we measured gene expressions by 3' SRP-RNA-Seq from RNAs HepaRG in 2D DMSO and in 3D GelMa culture at day 14 (Fig. 4A). As expected, genes coding for regulatory pathways of cell structural organization such as cell-substrate, extracellular structure organization and cell-cell adhesion are clearly up-regulated, reflecting their adaptation to the microenvironment, whereas processes mainly linked to cell metabolism and catabolic process of endogen (peroxisomal transport, isoprenoid metabolic process, fatty acid metabolic/catabolic process) and exogen (drug catabolic process, small molecule catabolic process) compounds, were down regulated compared to 2D culture by DMSO.

Then, we looked at the liver-specific gene expressions (LiGEP signature) as defined by Kim et al. [42]. This panel was based on the significantly differential RNA expressions between liver and non-liver samples. They developed an algorithm based on RNA-sequencing (RNA-seq) analysis to assess the differentiation or maturation status of 93 liver-specific genes validated by expression profiles in human Protein

Atlas database and by quantitative real-time PCR analysis. A modified version of this original list, excluding, inter alia, genes from xenobiotic metabolism, was used for transcriptomic analyzes (Supplemental Table 4). We added RNAs from freshly isolated primary human hepatocyte (PHH T0), used here as a standard reference. The expression profile of freshly isolated primary human hepatocytes is considered to be closest to the liver *in vivo* but cannot be reproduced *in vitro* due to the extremely rapid dedifferentiation of the human hepatocytes in 2D. Here, if a decrease in the expression of hepatic genes can be observed during the comparison of HepaRG GelMa and HepaRG 2D DMSO with the freshly isolated PHH, one notes a maintenance of the expression of hepatic genes of the LiGEP signature with a great homology of the expression profiles of HepaRG in GelMa and 2D DMSO, with both HepaRG in 2D and 3D showing a quite similar pattern of many hepatic gene expression (i.e. *ALB*, *ALDOB*, *SERPINA1*, *FGA*, *FGB*, *FGG*...) (Fig. 4B). By using real-time RT-qPCR, we confirmed that HepaRG cells in GelMa have a constant expression of various liver functions listed in the LiGEP heat map as well as receptors known to be involved in the biotransformation pathways of the drugs (i.e. *ALB*, *ALDOB*, *SERPINA1*, *HNF4A*, *NR1H4*, *NR1I2*), at short (7 days), medium (14 days) and long-term (28 days) after seeding (Fig. 5A). Their levels, throughout the 3D culture, were all superior to those of undifferentiated 2D HepaRG (2D) and were equal to or greater than those of cells in 2D DMSO conditions. We confirmed these results by analyzing the secretions of albumin and urea (Fig. 5B and C), which remained constant during the whole culture process. Moreover, the low expressions of *SOX9* and *CK19* could indicate that the cells in the GelMa could preferentially differentiate towards a hepatic lineage rather than a biliary lineage, contrary to the 2D cultures where the two lineages coexist after DMSO treatment (Fig. 5D) [29].

Concerning phase 1 biotransformation enzymes, the expression of these genes in 3D GelMa was closer to their expression in 2D DMSO than in PHH at T0 (Fig. 6A). By RT-qPCR, we validated that some mRNA expressions were equal (*CYP1A2*) or lower (*CYP3A4*, *CYP2B6*, *CYP2E1*, *CYP2C9*, *CYP2C19*) in 3D GelMa compared to 2D DMSO (Fig. 6B). In contrast, *CYP1A* and *1A2* activities could be induced in 3D cultures at the same level than in 2D DMSO (Fig. 6C). These induced activities can be observed all over the culture time allowing to analyze acute and chronic effects of the drugs in the 3D model until at least 28 days. *CYP3A4*, *CYP2B6* and *CYP1A2* activities could be induced by 2, 6, and 16-fold respectively indicating that 3D cells in GelMa respond well to all inducers (Fig. 6D and E). In the case of *CYP2B6*, activations were significantly higher in 3D than in the 2D DMSO cultures.

For phase 2 biotransformation enzymes and the transporters, as for phase 1 enzymes, the gene expression profiles in 3D GelMa were closer to those in 2D DMSO than in PHH T0 (supplemental Fig. 1A and 1B). As for the CYP mRNA expressions analyzed by RT-qPCR, expressions of the phase 2 enzymes and transporters were lower, equal to or greater in 3D compared to the 2D DMSO cultures, depending on the enzymes (supplemental figure 1C).

3.4. Evaluation of the fibrotic activity in micro engineered co-cultures

Based on these encouraging observations about the viability and functionality of hepatocytes cultures in 3D GelMa, we used this matrix to build topologically controlled 3D cocultures of different liver cell types, in order to design a model for evaluating the fibrotic process *in vitro*. For this purpose, we first built bio-printed micro-engineered co-cultures using HepaRG, LX-2 (derived from activated stellate cells [33]) and HUVECS (derived from umbilical endothelial cells) cells that could allow to obtain a model for evaluating the fibrotic process *in vitro*.

As a preliminary step, we analyzed the survival of LX-2 cells cultured in the GelMa (Fig. 7A). The WST1 activity remained stable or slightly increased during the first 14 days in the GelMa (Fig. 7B). Surprisingly, the LX-2 cells remained round in the matrix until at least 14 days of culture (Fig. 7C), whereas in collagen 1 matrix, they adopted a more elongated phenotype suggesting a transition to a myofibroblastic

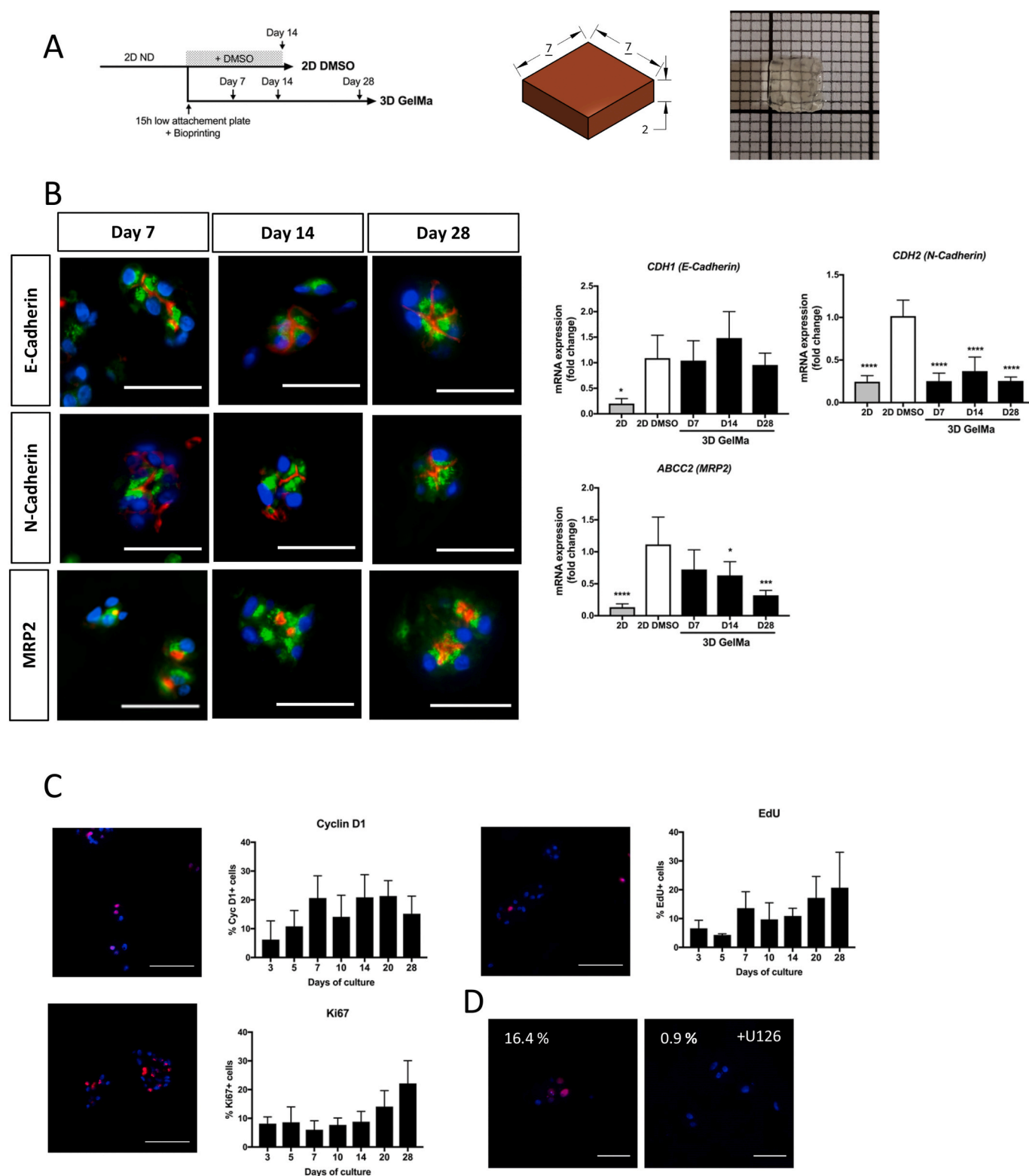


Fig. 3. Cluster organization, polarization and proliferation of the HepaRG cells in 5% GelMa. (A) Left: Experimental timeline for bioprinted HepaRG (3D GelMa) or 2D DMSO differentiated HepaRG (2D DMSO) Right: CAD design (size in mm) and macroscopic imaging of the printed structure. (B) Immunofluorescence imaging (left) and mRNA expression (right) of E-cadherin (CDH1), N-cadherin (CDH2) and MRP2 (ABCC2) at the indicated times after seeding. Green = albumin, blue = DAPI, Red = E-cadherin or N-cadherin or MRP2. Scale bar = 50 μ m. Significance levels of genes expression vs 2D DMSO results: * $p \leq 0.05$; *** $p \leq 0.001$; **** $p \leq 0.0001$. (C) Staining (left) and quantification (right) of proliferation marker Cyclin D1 and Ki67, and of the incorporation of the thymidine analogue EdU (DNA replication), in 3D HepaRG cells at the indicated times after seeding. Results are expressed as mean $\pm$ SD of $n = 3$ experiments. (D) Staining and quantification of EdU incorporation for 48 h in 3D HepaRG in the absence (left) or presence (right) of U0126 (MEK inhibitor). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

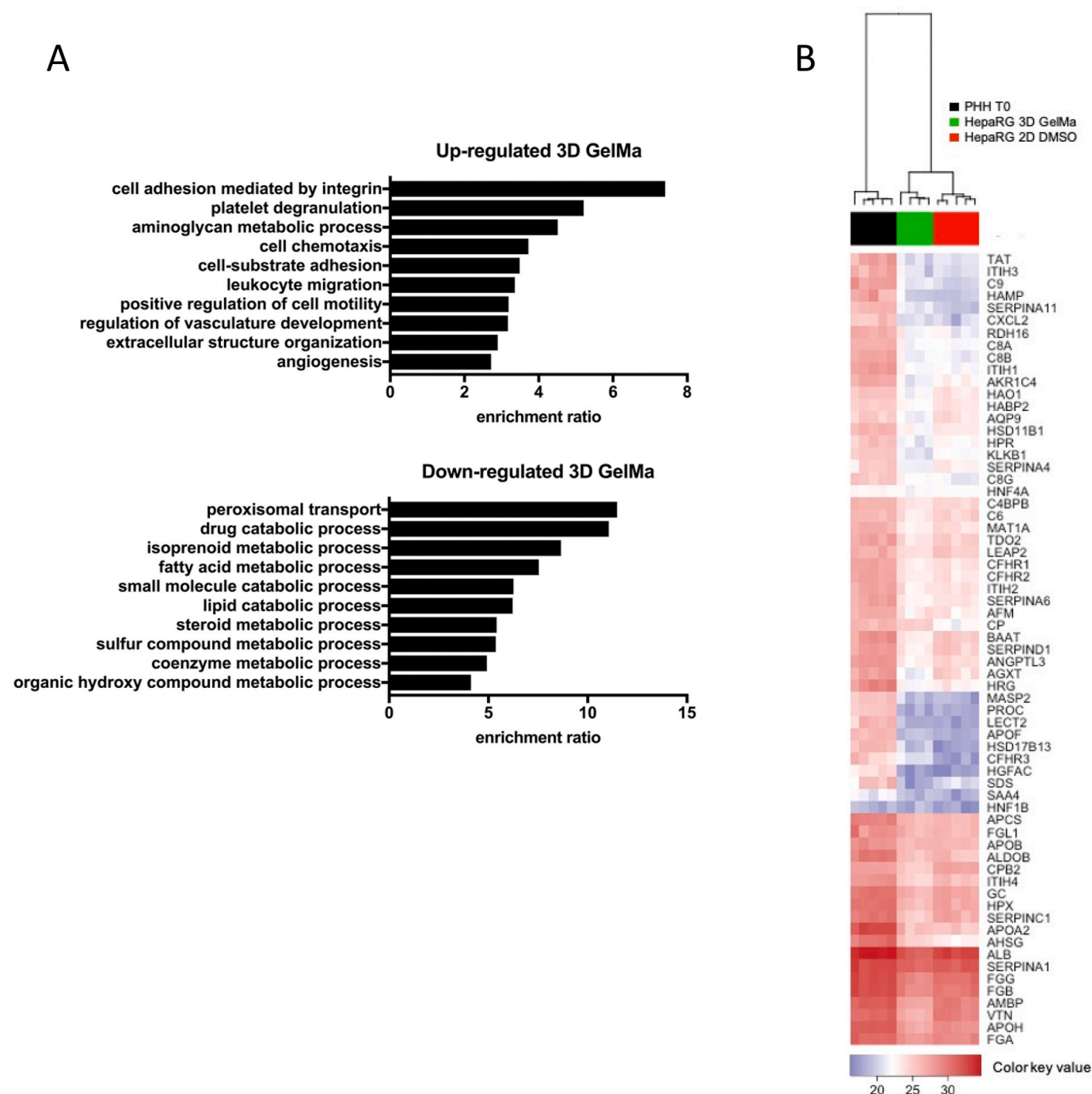


Fig. 4. Transcriptomic analysis of 3D HepaRG. (A) Gene set enrichment analysis based on the functional annotation of the differentially Up and Down-regulated genes in HepaRG cultured for 14 days in 3D in GelMa versus 2D in DMSO. (B) Heatmap ($-\log_2$ z-scores) of Ligep genes in freshly isolated human primary hepatocytes (PHH T0, black, 5 samples), HepaRG cultured in 2D with DMSO for 14 days (red, 5 samples) or HepaRG cultured for 14 days in 3D in GelMa (green, 4 samples). Samples and genes are hierarchically clustered based on Euclidian Distance according to their profile similarity. Elevated (red) and repressed (blue) expression are normalized to the mean of the gene expression. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

phenotype (Fig. 7C). We then assessed the capacity of LX-2 cell to respond to TGF β -1, a classical pro-fibrotic cytokine [43]. From day 2 of culture, cells were treated with vehicle (controls) or TGF β -1 (5 ng/mL every 48h) for two weeks (activation) or one week followed by an additional week without TGF β -1 (reversion) (Fig. 7A). In GelMa, expression of ACTA2, the major control of stellate cell activation, was clearly down regulated in control GelMa compared to cells cultivated on a 2D support (Fig. 7D).

After TGF β -1 treatment, the mRNA expressions of ACTA2, COL1A1, MMP2 and TIMP1, encoding enzymes or components involved in ECM homeostasis, were clearly increased in a time-dependent manner (Fig. 7E). These inductions were reversible and after TGF β -1 removal at day 9, the mRNA expressions at day 16 were significantly decreased and return to a level equivalent to that observed in the untreated control.

Then, we compared bioprinted HepaRG alone and a co-culture of HepaRG and LX-2 (CoC), within a suitable structure with large channels

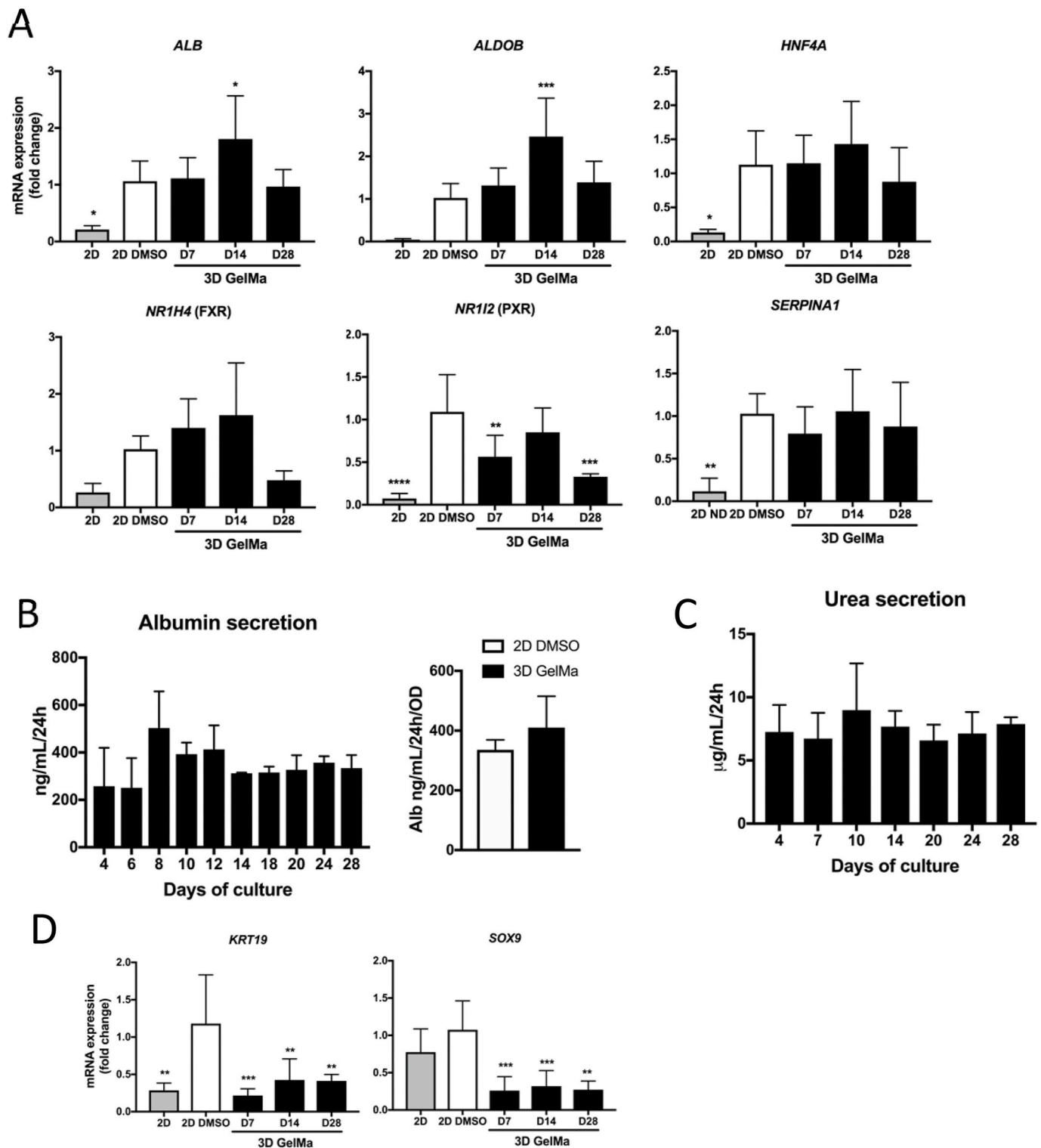


Fig. 5. Hepatic differentiation markers in HepaRG 3D GelMa. (A) Expression (RT-qPCR) of hepatic differentiation genes in 3D HepaRG (black) at days 7, 14 and 28 of culture, and 2D non differentiated HepaRG (2D), compared to 2D DMSO day 14 (white). (B) Kinetics (left) of albumin secretion by HepaRG 3D GelMa, and comparison (right) to 2D DMSO at day 14 of culture. (C) Urea secretion of HepaRG 3D GelMa over the culture time. (D) HepaRG in GelMa differentiate toward an hepatic rather than a biliary lineage: expressions of SOX9 and KRT19 in 3D GelMa at the indicated times (7, 14, 28 days) and 2D non differentiated HepaRG (2D) were measured by RT-qPCR and compared to 2D DMSO day 14 (white). All results are expressed as mean $\pm$ SD of $n = 3$ experiments. Significance levels of genes expression vs 2D DMSO results: * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$.

allowing solute/gas exchanges with the medium (Fig. 8A). HepaRG were bioprinted with large channels (Fig. 8B, red design of left panel) and the LX-2 were bioprinted inside the channels (Fig. 8B, blue) and, for both conditions, HUVECS cells were allowed to adhere and colonize the

entire surface of the hydrogel. HepaRG cells without HUVECS neither LX2 were bioprinted as controls. Expression of the green fluorescent protein (GFP) allowed us to visualize the homogeneous seeding of fluorescent HUVECS cells at the surface of the gel, 4 days after seeding

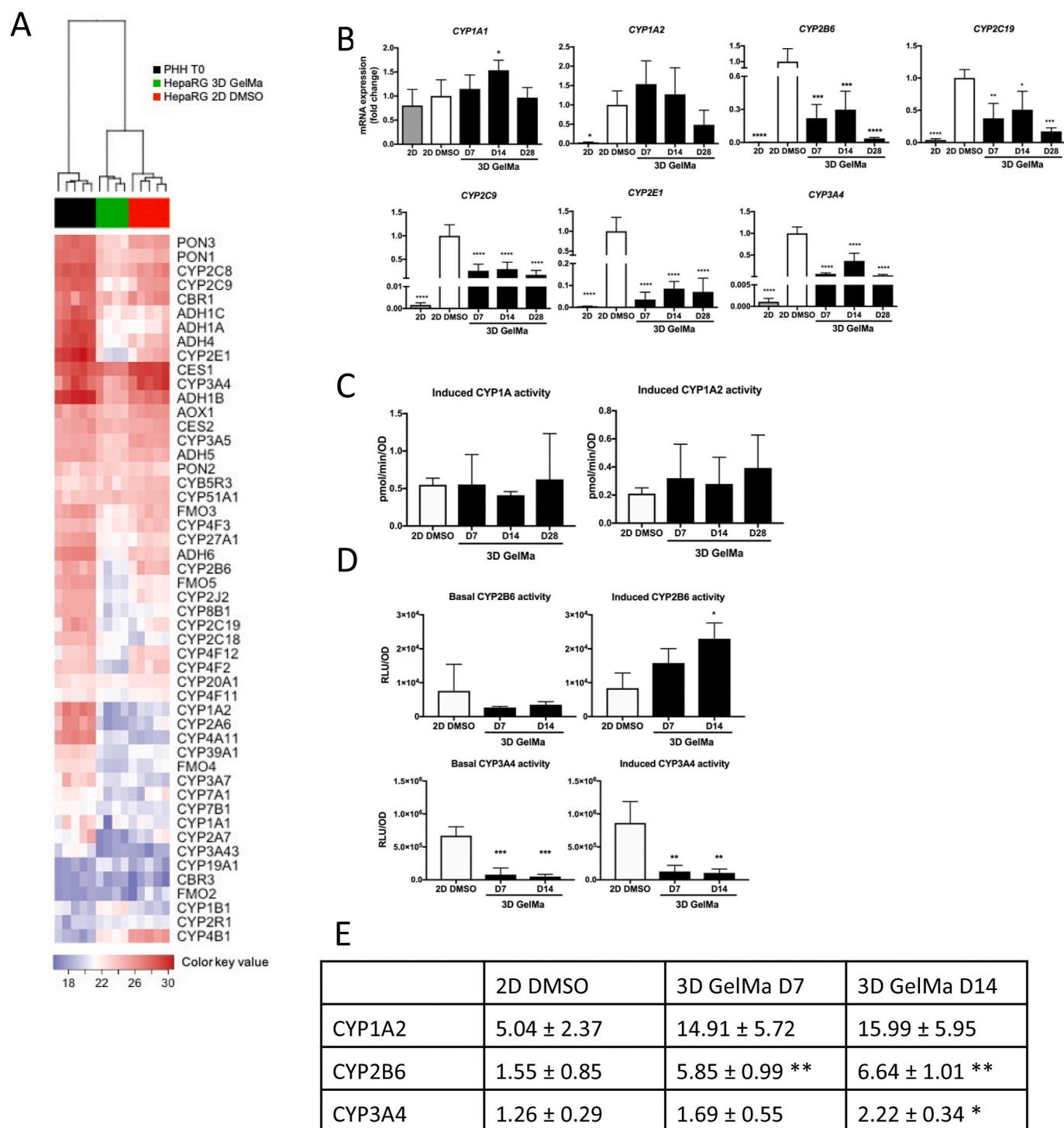


Fig. 6. Expressions and activities of phase 1 drug-metabolizing enzymes in the HepaRG cells cultured in GelMa. (A) Heatmap ($-\log_2$ z-scores) of phase I enzymes genes in freshly isolated human primary hepatocytes (PHH T0, black, 5 samples), HepaRG cultured in 2D with DMSO for 14 days (red, 5 samples) or HepaRG cultured in 3D in GelMa for 14 days (green, 4 samples). Samples and genes are hierarchically clustered based on Euclidian Distance according to their profile similarity. Elevated (red) and repressed (blue) expression are normalized to the mean of the gene expression. (B) Quantification of CYPs mRNA expression in 3D GelMa (black) at days 7, 14 and 28 of culture, compared to 2D DMSO at day 14 (white) or 2D undifferentiated HepaRG (2D; grey). (C) CYP1A and 1A2 activities of HepaRG induced by 3 MC treatment in 2D DMSO (white) and 3D GelMa (black) was assessed at day 7, 14 or 28 of culture by using the EROD/MROD assay (see M.M). Basal activity not detected. (D) CYP2B6 and 3A4 basal and induced activities in 2D DMSO (white) and 3D GelMa (black) were assessed at day 7 or 14 of culture by using Promega kit (see M.M). (E) Statistical analysis of CYPs induced activities fold increase over basal activities at 7 or 14 days of culture. Significance levels of genes expression or activities vs 2D DMSO: * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$; **** $p \leq 0.0001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

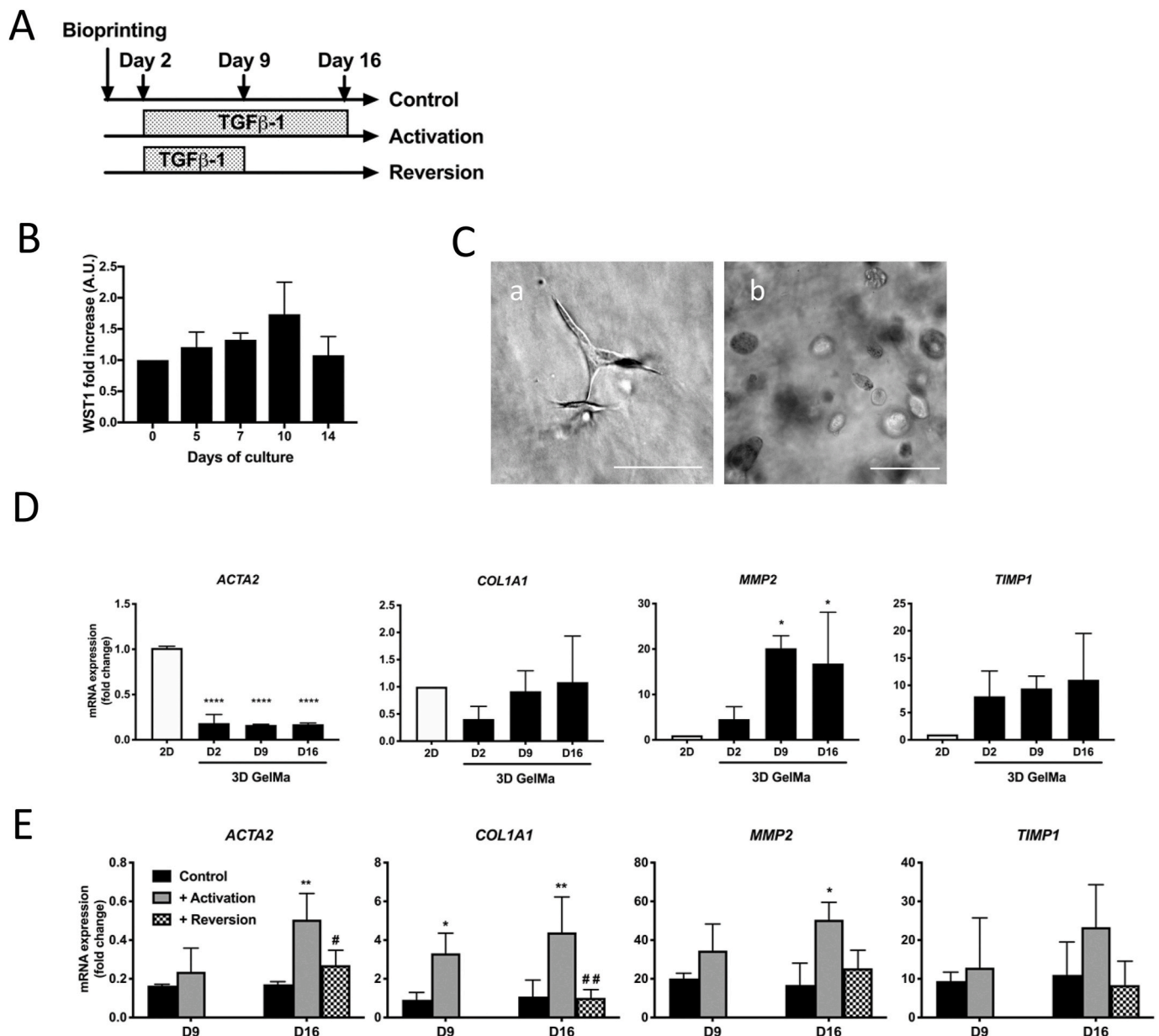


Fig. 7. LX-2 viability, cell morphology and mRNA expressions. (A) Experimental timeline for bioprinting and TGF β -1 mediated activation of LX-2. Structures were bioprinted and LX-2 activation was carried out by treatment with TGF β -1 (5 ng/mL every 48 h) for 14 days (activation). Alternatively, LX-2 were treated with TGF β -1 for 7 days, then cultured 7 days without TGF β -1 (reversion). (B) Viability of LX-2 cells cultured in GelMa 5%, LAP 0.1% for 14 days. Data are represented as fold increase of WST1 activities compared to that of control (Day 0). (C) White light imaging of morphology of LX-2 4 days after seeding in collagen 1 (a) or GelMa 5% (b) (scale bar = 50 μ m). (D) Quantification of ACTA2, COL1A1, MMP2 and TIMP1 mRNA expressions in 2D LX-2 at day 4 (white), compared to LX-2 at day 2, 9 and 16 of culture, in 3D GelMa (black). Significantly expressed genes vs 2D LX-2: * $p \leq 0.05$; **** $p \leq 0.0001$. (E) Quantification of ACTA2, COL1A1, MMP2 and TIMP1 mRNA expressions of LX-2 in GelMa (Control, black), after TGF β -1 mediated activation (grey) and TGF β -1 mediated reversion (squared) at the indicated times. Results were expressed as the n-fold ratios of target gene expression in samples to the mean expression values of the target gene in 2D cultivated LX-2 at D4. Significantly expressed genes vs control (*) or vs activation (#) at day 9 or 16: */# $p \leq 0.05$; **/# $p \leq 0.001$. All results are expressed as means $\pm$ SD of $n = 3$ experiments.

(Fig. 8B, right panel). The albumin secretion of the CoC, compared to that of HepaRG + HUVECs, showed that albumin secretion kinetics over time was the same in the two conditions, indicating that the coculture did not alter or modify the high functionality of the HepaRG cells in the bioprinted GelMa (Supplemental Figure 2). Although a slight decrease in *ALB* gene expression is observed in the HepaRG + HUVECs and CoC conditions (Fig. 8C), it can easily be explained by the presence, in these conditions, of non-hepatic RNA belonging to LX-2 or HUVECs cells.

After 14 days of culture, the three conditions (HepaRG, HepaRG + HUVECs, CoC) were activated or not by TGF β -1 for 7 days (Fig. 8A and B). When stimulated by TGF β -1, a significant loss of hepatic

functionality is reported by the fall in hepatic gene mRNA expression and albumin secretion (Fig. 8C) and an increase of genes involved in ECM homeostasis (Supplemental Figure 3). Nevertheless, no specific release of LDH (a cell death marker) in the medium was observed (Fig. 8C, right), and urea secretion remained stable over the 21 days of culture with or without TGF β -1 treatment (Supplemental Figure 4). When analyzing *COL1A1* expression, we showed that basal expression is significantly higher in CoC than in HepaRG and HepaRG + HUVECs, at a level close to those of LX-2 in GelMa alone (Fig. 8D, left). When activated by TGF β -1, all conditions showed a significant increase in *COL1A1* expression. This gene expression is correlated to pro-collagen 1 α 1

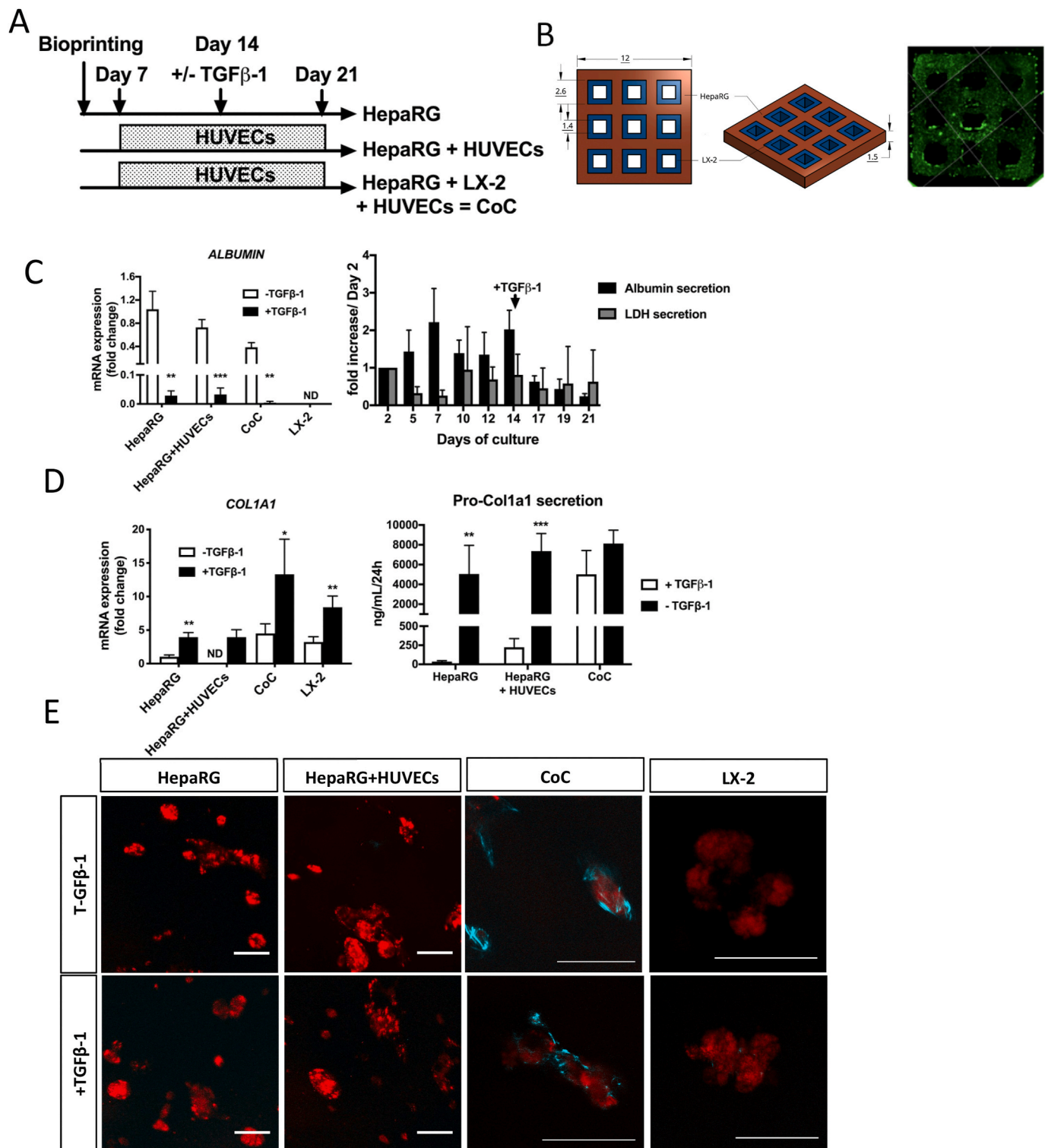


Fig. 8. The bioprinted 3D co-culture as a fibrotic model: micro-engineered co-culture made by bioprinting containing HepaRG, LX-2 and HUVEC cells. (A) Experimental timeline for bioprinting of HepaRG, HepaRG + LX-2 co-cultures and HUVEC cells seeding. At day 14, cultures were treated or not by TGFβ-1 5 ng/mL every 48h for 7 days. (B) Schematic representation of the coculture (CoC) model containing HepaRG (red) and LX-2 cells (blue) bioprinted in GelMa (size in mm). GFP-HUVEC cells have been added at day 7 of culture on the surface of the GelMa (Green, right picture: 4 days after HUVEC seeding). (C) Left: quantification of ALB mRNA expressions without (white) or with (black) TGFβ-1 in all 4 conditions. Significantly expressed genes vs HepaRG -TGFβ-1: **p ≤ 0.01; ***p ≤ 0.001. Right: quantification of albumin and LDH secretion of CoC + TGFβ-1 in media, expressed as fold increase of respective level at day 2. (D) Left: quantification of COL1A1 mRNA expressions without (white) or with (black) TGFβ-1 in all 4 conditions. Significantly expressed genes vs HepaRG -TGFβ-1: *p ≤ 0.05; **p ≤ 0.01. Right: quantification of procollagen1a1 (Pro-Col1a1) secretion in media at day 21, without (white) or with (black) TGFβ-1. Significantly expressed secretion vs D2: **p ≤ 0.01; ***p ≤ 0.001. (E) SHG/TPEF imaging of collagen deposition at day 21 (HepaRG, HepaRG + HUVECs, CoC) in all 4 culture conditions. Red = cells (auto-fluorescence), cyan = collagen 1 fibers (SHG). Scale bar = 50 μm. (HepaRG, HepaRG + HUVECs: 50 μm z-stack of images). All data are expressed as mean ± SD of n = 3 experiments. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

secretion in medium (Fig. 8D, right). Without TGF β -1 stimulation, almost none pro-collagen 1a1 can be detected in HepaRG or HepaRG + HUVECs, whether TGF β -1 treatment significantly induce its secretion. In CoC, a high basal secretion is measured at day 21 and no significant increase after TGF β -1 activation can be detected. The level of pro-collagen 1a1 secretion in CoC is close to the one of TGF β -1 activated HepaRG and HepaRG + HUVECs.

Our team has demonstrated the interest of second harmonic generation (SHG)/two-photon excitation microscopy (TPEF) as a tool to detect collagen fibrils *in vivo* and *in vitro* [8,44,45]. Briefly, SHG relies on the nonlinear optical interactions with non-centromeric fibrillar structures, such as collagen type I, allowing their detection and quantification in 3D tissues (Fig. 8E, cyan). TPEF is based on the ability of a near infrared light to excite endogenous fluorophores in cells in thick samples (red). As a negative control, the GelMa alone did not give any SHG signal (data not shown). We used this technology to assess whether pro-collagen 1a1 was correlated to further collagen fibril formation. SHG microscopy was used to analyze the fibrillar collagen deposition in all conditions. Interestingly, SHG microscopy allowed to detect collagen fibrils within the matrix only in cocultures of CoC cells, stimulated or not by the TGF β -1 (Fig. 8E). We also confirmed that no collagen deposition could be observed in HepaRG cultured alone or with HUVECs, and, unexpectedly, activated or not by TGF β -1. In such an interesting way, no collagen fibrils were ever observed after induction of the LX-2 cells by TGF β -1 in our bioprinted GelMa model, when LX-2 cells were cultured alone.

4. Discussion

The main goal of this study was to bio-manufacture an adequate 3D cell/matrix model which could allow long-term differentiation of HepaRG cells and would serve as a base for further co-culture and microfluidic developments. Moreover, we have implemented an innovative technology to finely control the topology of these co-cultures. As a first step, we aimed to improve the 3D culture of the HepaRG cell line in GelMa, which promote cell-cell and cell-ECM interactions, characteristics that are essential for improving or maintaining cell functionality in the long term.

Very few studies, so far, have been able to develop an efficient and long-term viable 3D bioprinted hepatic model. One of the main obstacles is, among others, the long and critical process of bio-ink selection and determination of the parameters for its efficient use. Here, we used GelMa, a versatile and widely ink used in 3D bioprinting [18,27,41], for the development of a 3D liver model based on the encapsulation of the HepaRG cell line. So far, the use of this bio-ink for the bioprinting of hepatocytes has been limited to 10% w/v GelMa matrices [18,19,41] while we have chosen to decrease the concentrations up to 5% GelMa, the lowest concentration which showed bioprinting capacities. This choice was motivated by the fact that it has been previously well demonstrated that the decrease in matrix concentration, as well as its substitution rate, are correlated with a decrease in the stiffness of the gels [19,46,47]. The increasing mechanical stresses associated with this rigidity are known to negatively influence the proliferation and viability of the encapsulated cells. Likewise, the concentration of GelMa and its degree of crosslinking have been shown to be inversely proportional to the degree of porosity and to the size of the pores obtained [46]. These pores, by ensuring the transport of nutrients and oxygen to the cells, are important actors in the survival of the encapsulated cells. The formation of a network of large pores due to a reduced degree of methacrylation is also more permissive for the proliferation, the organization and the migration of the encapsulated cells [46]. The LAP concentration, lighting duration and intensity has been shown to be proportionally linked to an increase in photo-induced crosslinking, and therefore in the stiffness of the gels [47]. Here, by reducing it to the minimum useable (0.1%, 1 min, respectively), we considerably improved the viability of the cells over time, without losing structural stability of the printed structures.

This also allowed us to improve cell viability by limiting exposition to free radicals produced during radical photopolymerization [40] and by limiting the number of cross-linked methacrylate substitutes, thus allowing to finally tune the rigidity of 3D structures. These optimized parameters lead to a physical environment allowing us to achieve long term 3D cultures of bioprinted Huh7 and, for the first time, of the HepaRG cells line.

This culture, over a period of at least 28 days, of HepaRG cells is a major first step for obtaining bioprinted liver models competent for hepatotoxicity or ADME studies. Indeed, other existing bioprinted models are either based on the use of hepatic cell lines such as HepG2 or L02 [19,41,48], which are less metabolically competent and differentiated, or murine hepatic cells [22,23], which does not fully recapitulate human metabolism. HepaRG is a unique cell line which, due to its high metabolic capacities once fully differentiated in DMSO 2%, is currently the “gold standard” for xenobiotic metabolism and hepatotoxicity tests [29,49]. So far, this cell line has been successfully printed in either GelMa [18] or a blend of ECM/Alginate/GelMa [20] bio-inks. In these previous models, the cells are, at best, viable over 14 days, a time too short to study chronic treatment toxicity, a main limitation for chronic toxicity evaluation assays. Here, we could demonstrate continuous viability for at least 4 weeks. Also, in our 3D model, cells self-organize into polarized structures, mimicking the hepatocyte polarization *in-vivo*, as shown by the specific location of E Cadherin and MRP2. The establishment of a cell architecture that recapitulates E-Cadherin-dependent interactions is probably one of the factors for the maintenance of long term hepatic functionality in our models, as E-Cadherin is known to inhibit *in vivo* signaling pathways inducing hepatocyte dedifferentiation [50,51]. We assume that the re-creation of these cell-cell interactions is due, as in many 3D hepatic models [8,9], to their culture in a 3D matrix environment whose stiffness is closer to that of a healthy liver (± 4.5 kPa), as compared to 2D cultures on plastic plates ($\sim 1 \times 10^7$ kPa) [52]. Furthermore, early interactions between cells are also critical for the long-term differentiation of the HepaRG cells in GelMa. Indeed, HepaRG must be seeded transiently on low attachment plate before being embedded into the Gel-MA matrix. In absence of these early cell interactions enhancement, the HepaRG cannot fully differentiated in GelMa (data not shown).

The HepaRG cell line shows great plasticity in 2D culture, being able to differentiate to both biliary- and hepatocyte-like cells [53]. After seeding at low density, the cells actively proliferate until they reach confluency and differentiate toward these two lineages. A maximum of differentiation is reached after 2 weeks exposure to DMSO, and hepatocyte-like cells represent at least 50% of the cell population. Our data on HepaRG cells in GelMa, cultivated without DMSO showed survival and hepatic differentiation close or higher to the cells cultured in 2D with DMSO. The differentiation was optimal at 14 days of culture and stayed at a high level for up to 28 days. Noteworthy, HepaRG bioprinted in GelMa seemed to differentiate preferentially toward a hepatic lineage, although the exact phenotype must be refined by extended analysis of immunohistochemical and transcriptomic profiles. This observation is in line with previous studies showing that HepaRG cells encapsulated in 3D matrices form hepatocyte-like colonies rather than cholangiocytes [54, 55].

A major interest of developing 3D hepatic models is to use them as an *in vitro* alternative to screen the hepatotoxicity and xenobiotics metabolism. This implies that the metabolic capacities of these models must be as close as possible to those observed *in vivo*. However, with the exception of the ExVive 3D model [24,25], few bio-printed liver models have been evaluated based on these capacities. Bioprinted structures based on HepaRG cultures are constrained by the need to cultivate those cells for 2 weeks in DMSO to reach optimal differentiation, removal of DMSO from the medium resulting in a quick fall in hepatic differentiation markers [29]. This is why actual HepaRG bioprinted models are based, so far, on the encapsulation of DMSO pre-differentiated cells on the matrix before bioprinting [18,20,56]. Moreover, the quantification

of the metabolic capacities of the models are limited, based solely on the evaluation of the expression or activity of the sole CYP3A4, without a significant improvement in metabolic activities compared to cells in 2D [20]. Thus, their use as hepatic models for xenobiotic metabolism has so far been extremely limited. In an interesting way, 3D cultured hepatocytes in GelMa were able to differentiate in the absence of DMSO. We showed a nearly equivalent expression of the biotransformation enzymes of phase 1 as compared to the 2D DMSO cultures, although their activities (CYP1A, 1A2, 2B6, 3A4) could be induced throughout the 3D culture at a higher rate. We thus demonstrate, compared to pre-existing models, the capacity of bioprinted HepaRG in GelMa to ensure, in the short, medium and long term (7, 14 and 28 days of culture), the biotransformation of xenobiotics at levels equal to or greater than fully 2D differentiated cells. This will allow to study the effects of chronic exposure to drugs and environmental products, in the absence of DMSO, which is a powerful reactive oxygen species producer and inductor of numerous CYPs, including the CYP3A4 and CYP2B6.

Another strength of our model, never demonstrated so far, is the ability of the differentiated HepaRG cells to constantly proliferate throughout the culture in GelMa. Thus, 3D cultures of HepaRG in GelMa overcome a major limitation of 2D cultures where proliferation and differentiation are mutually exclusive. These proliferative and differentiated properties of the HepaRG cells will be of great interest for studies focusing on the genotoxicity induced by drugs and environmental contaminants that need to be metabolized by the liver to produce genotoxic and/or mutagenic products [57].

Models where different liver cell types are co-cultured in 3D partially recreate the complex interactions between parenchymal and non-parenchymal cells, allowing to explore physiological pathways in a context closer to that of *in vivo* organs [32,34,35,58]. We demonstrated that the bioprinted co-culture of HepaRG/LX-2/HUVECS in GelMa constitutes a suitable model for studying collagen synthesis and deposition, emphasizing the major role of regulation between parenchymal and non-parenchymal cells in the collagen deposition [12]. This might be a major advance for studying hepatic fibrosis, as there are currently few *in vitro* physiological models of this pathology. Indeed, such models have to deal with several constraints, among them the need of putting together different cell types involved in hepatic fibrosis, the ability of stellate cells to activate towards a myofibroblastic phenotype, and a 3D environment allowing the ensuing progressive deposition of the ECM [25,32,59]. The reaction of HepaRG cells to TGF β -1, whether in mono or co-culture, could mimic that found in the progressive development of fibrosis *in vivo*. Indeed, during this pathology, the TGF β -1 is a promoter of fibroblastic/migratory phenotype. *In vitro*, the dedifferentiation of human hepatocytes was also objectified by the decrease of the expression of *ALB* and *HNF4A* after TGF β -1 treatment [60]. For the stellate cells model, we present, as already described by Norona et al., the maintenance for several weeks, of the LX-2 cell line (a cell line obtained from activated human primary stellate cells), in a “quiescent”-like phenotype. Unlike LX-2 cultured in 2D, the expression and synthesis of two markers of the activated HSC phenotype, respectively *ACTA2* and type I collagen, are strongly inhibited when the cells are embedded in a GelMa matrix, confirming the capacity of a flexible matrix to support the quiescence of stellate cells *in vitro* [25,35,61]. As *in vivo*, the LX-2 cells remain however capable to respond to pro-fibrosing molecules such as TGF β -1, and, strikingly, to reverse to a “quiescent”-like phenotype upon withdrawal of this pro-fibrotic cytokine. It must be noted that the reversibility of this activation, as seen in our model, is also described *in vivo* [62,63], and in *in vitro* murine models [35] but has yet to be precisely described in an *in vitro* human model [61]. LX-2 cells cultured in GelMa thus offer new perspectives for the study of cellular mechanisms linked to the activation and reversion of HSCs, in the context of human hepatic fibrosis progression and reversion.

Here, we show that activation by TGF β -1 of HepaRG, HepaRG + HUVECs, LX-2 and CoC cells strongly induces the expression of the *COL1A1* gene, and the secretion by these cells of pro-collagen1a1.

Noteworthy, although the use of SHG biphotonic microscopy did not evidence any deposit in presence of TGF β -1 of collagen fibers, a major marker of the development of hepatic fibrosis, in monocultures of LX-2, HepaRG, HepaRG + HUVECs cells. This deposit was detectable only in co-culture with HepaRG, LX-2 and endothelial cells even in absence of TGF β -1. This illustrates and confirms the fact that the development of fibrosis is a complex phenomenon, strongly based on cell communication between the different cell types. Thus, the *in vitro* secretion of matrix proteins such as pro-collagen 1a1, as well as the synthesis of matrix remodeling factors, which can be detected *in vitro* in 2D mono-cultures of hepatocytes [64] or HSCs, might not be translated by a further organization in fibrillar structure, thus lacking a critical step for mimicking *in vivo* liver micro-environment.

The article presented here focuses on determining the parameters for using a 3D bioprinting matrix, the GelMa, which allows 3 things: 1/ physico-mechanical parameters allowing it to be used in 3D bioprinting, as shown by the impression of two different models, in mono and coculture, 2/the DMSO-free differentiation, in this matrix, of HepaRG cells towards a hepatic phenotype by the microenvironment 3/the maintenance of stellate cells, here LX-2, at a “quiescent stage” that can be activated thanks to these same conditions. The results obtained in our study confirm the importance of sophisticated interactions between the different hepatic cell types to obtain a deposit of collagen and to build relevant pathological models. This further illustrates the need for developing topologically controlled 3D co-culture models to recapitulate the progression and reversion of liver fibrosis. Future studies will expand the genomic and histologic characteristics of these bioprinted fibrosis models in the presence of pro- and anti-fibrotic drugs. Finally, coupling bioprinting techniques with perfusion-enabled channel systems combining HepaRG and human stellate cells will be developed in the future to improve organ-on-chip applications for pharmaceutical developments. Likewise, the use of 3D bioprinting, under our conditions, for the creation of chips coupled with microfluidic systems seem to us to be promising options.

Funding

Financial support: Supported by the Institut National de la Santé et de la Recherche Médicale (Inserm), University of Rennes 1, INCA #2018-142, Ligue contre le cancer du Grand Ouest. MC has been a recipient of a fellowship from Inserm and the Région Bretagne.

Credit author statement

Marie Cuvellier: Investigation, Writing – original draft, review & editing; **Frédéric Ezan:** Investigation; **Hugo Oliveira:** Investigation, Resources, Writing – review & editing; **Sophie Rose:** Resources, Methodology; **Jean-Christophe Fricain:** Supervision; **Sophie Langouët:** Conceptualization; **Vincent Legagneux:** Formal analysis, Writing – review & editing; **Georges Baffet:** Conceptualization, Funding acquisition, Writing – original draft, review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank the Dr Nathalie Theret (IRSET) for giving us the LX-2 cell line, the ART BioPrint (INSERM, Bordeaux) and the platforms from the SFR Biosit (UMS CNRS 3480/US INSERM 018, Biogenouest, University of Rennes 1): Microfluidic & Bioprinting Facility of Biosit, MRic-Photonics, MRic-Biphoton, H₂P₂, CRB Santé de Rennes.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biomaterials.2020.120611>.

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